Can Willingness to Pay be Identified without Price Variation?

What Big Data on Usage Tracking Can (and Cannot) Tell Us

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Abstract

We study how to obtain consumers willingness to pay (WTP) for subscription products, where consumers pay a monthly fixed price. We show that variation in usage and subscription choice together can identify the elasticities and the WTP distribution. We propose a novel estimation strategy to recover the willingness to pay distribution in the absence of price variation. In addition, we show how price variation can help identify the functional form in which the usage affects the WTP. We demonstrate an application of our method with both usage and monthly subscription data from a music streaming service. We recover the conditional distribution of WTP based on demographic variables, and find positive age elasticity of the WTP, and that male subscribers are willing to pay more for the service. We estimate the demand curve for the monthly plan and the distribution of WTP for a counterfactual subscription plan, and we determine how to optimally set prices for the new plan.

1 Introduction

The feasibility of demand estimation in economics and marketing has depended on the presence of data with price variation. However, we observe many different markets where consumers subscribe to a limited time frame (e.g. monthly) plan that provides some maximum allowed consumption in that period. To identify how consumers value various features

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of the product or service, the firm needs to be able to recover the consumers' willingness to pay (WTP) or valuation distribution. Identifying the mean and more generally the entire distribution of WTP, given consumer and product characteristics is a first step required to understand and predict demand responses. Managers can then use these estimates to identify how alternative products should be priced. Consider the example of Netflix, which has a monthly plan for $9.99 in the US. Suppose the firm is interested in evaluating how demand might vary with price increases, or identifying how a weekly plan might impact valuations and profits, we would need to obtain the WTP distribution.

Subscriptions are becoming increasingly popular across the world for both physical and digital products and services with growth over 100% over 2013-2018 (**??**). There are a number of reasons for this popularity, including low marginal costs (relative to fixed costs), reducing consumer risk, avoiding transaction costs, providing predictability in revenue stream as well as increasing loyalty (**??**). Subscription plans are prevalent across a wide variety of industries, ranging from media to software-as-a-service (SaaS) to eCommerce and transportation, as detailed in **??**.

In most subscription commerce markets detailed here, price variation is fairly rare or non-existent, except for occasional promotions (free trial). As an example, Spotify has always set the monthly price for unlimited ad-free streaming to be around $10. The absence of price variation poses a challenge for using the common revealed preference approach to recover the distribution of the WTP, which relies on price variation, a feature common to the entire literature (**????**). Firms in such markets set these prices based on market research typically using conjoint or similar survey elicitation responses (**??**). While conjoint is a very useful tool to obtain relative preferences, it has difficulty in capturing accurately valuations that are further away from the market price, and consumers have been found to have a different WTP when making actual purchase choices.

When prices do not vary, the key insight of this paper is to recognize there may be variation in other elements of the data that can be leveraged to obtain the distribution of WTP. In subscription settings, firms typically observe consumers churn or leave and then return (which happens commonly with Netflix). In addition, the firm observes microdata on the product usage or consumption, as well as rich data on a variety of consumer and product characteristics. Often these big data are available at high velocity or frequency, especially for digital products. Below we will use the usage and consumption interchangeably. Given this background, we examine the following research questions:

(i) In a subscription usage setting without price variation, what can we infer about the distribution of consumer valuations? If price does not vary, what variation in the data is required to obtain any such inference?

Table 1: Subscription Plans

Industry	Product or Service	Price ($)	Period	Total subscribers
Media & Entertainment	Netflix	9.99	Monthly	23 million (US)
	Spotify	9.99	Monthly	70 million (World)
	New York Times	3.75	Weekly	4 million (US)
	MoviePass	19.95	Monthly	2 million
	Kindle Unlimited	9.99	Monthly	–
	Apple News	9.99	Monthly	36 million
Software-as-a-Service	Microsoft Office 365	9.99	Monthly	120 million
	Adobe Creative Cloud (One App)	20.99	Monthly	15 million
	Dropbox Premium	9.99	Monthly	>11 million
Membership Clubs	Costco (Basic)*	60	Annual	94 million
	Amazon Prime	119	Annual	90 million
	24 hour fitness (Gym)	40	Monthly	4 million
eCommerce	Harry's	35	Monthly	–
	Birchbox	15	Monthly	2 million
	Rent the Runway	159	Monthly	6 million
Transportation	Public Transit Pass (MTA)	121	30-days	–
	Uber Ride Pass*	14.99	Monthly	–
	Jetblue "All You can Jet" Pass	699	Monthly	–

(ii) In settings with price variation, what additional inference is possible?

(iii) Can we determine the impact of counterfactual product and pricing choices by the firm?

(iv) How can the firm measure the impact on valuations of introducing product improvements?

The main contribution of this paper is to propose a novel method to identify and estimate semiparametrically the conditional distribution of the WTP given product features and customer characteristics *when price variation is absent*. To the best of our knowledge, there is no research that demonstrates how to obtain the WTP distribution in the absence of price variation. We then demonstrate the practical value of obtaining these estimates.

The key intuition for our result is that even though price is identical among customers, there is still variation in the price of *per unit* consumption when there is variation of consumption (usage) among users. The variation in usage or consumption along with *either*

temporal or cross-sectional variation in subscription choice together can help identify the conditional distribution of the WTP under a set of reasonable assumptions. Using a dataset provided by a music streaming service company, we estimate the conditional distribution of WTP and elasticities of the WTP for its monthly music streaming service plan to demographic characteristics. We find positive elasticity of the WTP with age, and find male subscribers are willing to pay more for the plan. We also estimate the mean of log WTP for different age and gender groups. We estimate the demand curve for the monthly plan and the distribution of the WTP for a counterfactual weekly subscription plan.

The crucial aspect required for our method is the separation of purchase (subscription) and consumption (usage). The separation implies that two consumers typically have different amount of consumption if they have both subscribed. Observe that this holds naturally in subscription settings like those in **??**. In addition, *even in typical packaged goods*, there is a separation between purchase and consumption, but on most such cases we do not observe the consumption. However, if consumption data were observable, our approach would be applicable to such settings too. With the advent of technological advances likes the 5G telecommunications and the Internet of Things, the measurement of such consumption is likely to become more prevalent in the future. In fact, there are some companies that already offer such services, notably LG has a smart fridge that monitors consumption of perishables like milk with the idea that these could be automatically replenished without direct consumer intervention.[1]

There are a wide variety of product and pricing choices where the proposed approach would be useful. First, the firm might want to assess the impact of improvements in product quality or other product characteristics on the distribution of consumer valuation. Second, a firm might consider changing the subscription period. For example, introducing a weekly plan or an annual plan in place of a monthly plan. Third, the firm might want to charge a price for each unit consumed in addition to the subscription price, making it a two-part tariff. Fourth, the firm might evaluate introducing quotas or quantity restrictions. Such restrictions are often imposed in otherwise unlimited plans for mobile data service and cable plans. Netflix and Comcast have been reported to throttle some heavy users.[2]

The rest of the paper is organized as follows. **??** reviews the literature. In **??**, we show how to identify the conditional distribution of the WTP and the conditional mean of the log WTP given product and consumer characteristics, which can be used to derive the demand curve and the elasticities of the WTP. **??** studies the role of price variation in identifying the WTP distribution. In **??**, we discuss the estimation details. In **??**, we show the relevance of

[1] See for example: **?**

[2] See for example: **?**

our method in empirical research by estimating the WTP distribution for a music streaming service. The appendices contain technical proofs and Monte Carlo studies of the performance of our method.

2 Literature

There are multiple streams of literature focused on measuring and characterizing WTP or the distribution of valuations. See **?** for an excellent overview. There are a few different approaches to eliciting WTP, either at an individual level or in obtaining a market-level aggregate. An important distinction should be made between methods that use stated preference to obtain hypothetical WTP, and that use revealed preference to obtain real WTP. In the real WTP case that involves consumer choice, to the best of our knowledge, there is currently no general method that can obtain the WTP distribution in the absence of price variation. On this point lies the primary contribution of this paper.

Within the *stated preference* stream of literature, customer populations are surveyed to obtain an estimate of WTP. Such approaches are typically used to obtain hypothetical WTP since consumers do not have to actually pay a price or face financial consequences. These include direct surveys of consumers or buyers, which remains used in contingent valuation type settings without product variation. For example, consumers might be asked how much they value a particular public and environmental goods like a park (**??**). The appeal of this methodology is in its simplicity and in obtaining an economically relevant quantity, although researchers have long pointed out the challenges in obtaining an accurate estimate (**?**).

Another method that was developed in marketing and has developed a strong stream is the indirect conjoint analysis method (**??**). See (**?**) for a comprehensive perspective. Conjoint guides consumers into making rank-ordered preferences from a limited choice set. With sufficient observations, it is possible to obtain an individual level willingness to pay not just for the overall good, but its constituent features, e.g. battery life in a device.

There are many typical advantages of this approach. First, they are relatively easy to implement, and allow the researcher (or firm) to introduce exogenous variation in product characteristics or attributes, prices as well as in choice sets available to consumers. These sources of exogenous variation are powerful in providing clear identification through induced variation. Another advantage is that they can be used to test how the market values hypothetical improvements or changes in advance of actually making them. Within this stream there are two broad approaches: direct surveys and choice-based conjoint. Direct surveys ask individuals to place a monetary value on a product or service (contingent valuation). Conjoint on the other hand asks consumers to rank order choices, which can vary based on

price as well as other characteristics. A disadvantage of conjoint is that it is usually set at market prices, implying that with higher dispersion of willingness to pay, valuations that are further away from the market price will not be accurately captured.

Next is the well-established literature on demand estimation *using observational data*, either at the market-level (**??**) or micro-data based on individual consumers like in much of the marketing literature (**?**). In these cases, the idea of price variation is central to identification (see for example **???**). In addition, endogeneity is often an important concern in demand estimation (**?**). The typical issue is that prices are set based on unobservable characteristics of products or based on market or consumer characteristics. Thus, we cannot rely on exogenous variation to be able to identify demand. We must therefore rely on instrumental variables or another approach like control function to identify demand and WTP. While this problem has been well recognized, it is not just a theoretical concern. Since the time of **?**, it has been noted that without carefully accounting for unobserved characteristics, positive price coefficients can be obtained, implying consumers prefer to pay more, all else equal.

Another set of papers involve field experiments, where researchers have carried out different experimental designs to elicit the WTP distribution. These involve either an auction based approach (Vickrey auction) as in **?** or involve a stochastic price generation mechanism (**?**) that induces incentive compatibility among the participants.

Within marketing, there is a rich stream of literature focusing on specifying and estimating rich models of consumers heterogeneity. These models either use random coefficients for individual households or a hierarchical Bayesian approach, and help in designing and evaluating targeted interventions to specific households (**?**). This focus on individual heterogeneity is very helpful in targeting promotions (e.g. coupons) at the individual or household level, allowing the more efficient generation and capture of surplus by the firm. The present paper shares many features with this stream in the sense that we are interested in characterizing the valuation distribution at the individual customer level, and potentially condition it on observable demographic characteristics.

However, there is the key question of whether stated preferences correlate with actual behavior or revealed preferences. Several studies have found significant differences in elicited valuation depending on the specifics of the method used to obtain it. As detailed across a variety of studies, the stated or hypothetical WTP is often found to be higher than revealed WTP (**???**).

It is striking that none of the methods provide any help when there is no price variation in the data or under an experiment. Even the use of instruments is challenging in such a case because there is no instrument that can be correlated with a constant price, and be

uncorrelated with the unobserved errors.

The most related paper is **?**, who estimate demand for residential broadband using usage (download/upload in GBs) and plan choice (e.g., unlimited usage plans vs usage-based plans) data when subscribers face a three-part tariff. Under this three-part tariff, subscribers pay a fixed fee each month that pays for all usage up to certain allowance, and they will be charged at an overage price for each GB of usage in excess of the allowance. A forward looking user would realize that the cost of today's usage even before hitting the allowance threshold depends on how much one has used up to today. As a result, even though the fixed fee and overage price are identical for all subscribers of the same cable company, there is still variation in the shadow price of usage on day t, because there is variation in the amount of usage up to day t. So their identification strategy also explores the variation of individual usage and high frequency usage data (e.g., daily usage within a billing month). Using this observation, they identify and estimate consumer's utility function and evaluate various counterfactual policies about broadband.

The key difference from **?** is that our identification arguments do not rely on three-part tariff that creates the variation of shadow price of usage as the accumulated usage approaches to allowance. This is relevant in practice because many technology products do not use three-part tariff pricing model (e.g., music streaming service like Spotify in our empirical application). Our identification strategy explores the information contained in both usage and the subscription choice (subscribe or don't subscribe). We demonstrate that together with usage data, subscription decisions contain substantial amount of information that could help identify the WTP without further variation or additional modeling assumptions. We also explore the limitation in identifying the WTP of observing only usage and subscription choices without price variation, and show the additional value of price variation.

3 Identify the WTP from Usage Data without Price Variation

3.1 Model setup and parameters of interest

To make the exposition concrete, we use a monthly gym membership, as an example application. Of course, our method applies to many other similar markets, such as online music streaming service as in our empirical application. Consumer i decides whether or not to buy a monthly gym membership at price P that is identical for each consumer in the market and constant over time. Subscribers can have unlimited usage once they purchase the

plan. Suppose we have a random sample of n individuals from the population of interest. The population of interest can be all individuals who have ever subscribed the plan or all individuals in one region who *may or may not* be a customer of the gym. Depending on the definition of the population of interest, we will only be able to identify the distribution of the WTP in that specific population.

Let $S_i \in \{0\,(\text{no}), 1\,(\text{yes})\}$ denote the subscription choice—$S_i = 1$ if individual i have the subscription. We adopt a money-metric representation of expected utility of the service. Let

$$W_i - P$$

be the expected utility of the service, if i subscribed. Here W_i is her WTP. If she did not subscribe, there is an outside option (e.g., running) for her, and the money-metric expected utility of this outside option equals μ. She will subscribe if and only if she believes that the expected utility of the service, $W_i - P$, is greater than the expected utility of the outside option, μ, hence

$$S_i = \mathbb{I}(W_i - P > \mu) = \mathbb{I}(\ln W_i > \ln(P + \mu)),$$

where $\mathbb{I}(\cdot)$ denotes an indicator function. In the absence of price variation, W_i is at most identified up to a constant when μ is unknown.[3] For example, adding a constant to both W_i and μ does not affect observed consumer's choice. In this section, we normalize $\mu = 0$. We will discuss how to identify μ later using (limited) price variation.

Let X_i denote a vector of observable consumer characteristics (e.g., sex, age, and distance to the gym) and product features (e.g., number of fitness lessons and swimming pool availability). The WTP W_i is a function of X_i.

We will show how to estimate the elasticities of the WTP to changes in X_i, and in general the conditional cumulative distribution function (CDF) $F_W(W_i \mid X_i)$ of W_i given X_i. The elasticity of the WTP to the change of a continuous X_{ji} (e.g., distance to the gym) is

$$\mathrm{E}\left(\frac{\partial \mathrm{E}(\ln W_i \mid X_i)}{\partial X_{ji}}\right),$$

For a dummy (or categorical) variable X_{Ki} (e.g. whether or not the gym has a swimming pool), the elasticity is

$$\mathrm{E}[\mathrm{E}(\ln W_i \mid X_{1i}, \ldots, X_{K-1,i}, X_{Ki} = 1) - \mathrm{E}(\ln W_i \mid X_{1i}, \ldots, X_{K-1,i}, X_{Ki} = 0)].$$

In both formulas, the outer expectation is taken over X_i in the population. These elasticities are relevant for product design. The CDF $F_W(W_i \mid X_i)$ can be used to identify the demand

[3] It will be clear later that this is one example of what usage tracking alone cannot tell us.

curve. First, one can obtain the unconditional distribution $F_W(W_i) = \mathrm{E}(F_W(W_i \mid X_i))$. Suppose p^c is a counterfactual price, we know that

$$D(p^c) \equiv \Pr(S_i = 1 \mid P = p^c) = \Pr(W_i > p^c) = 1 - F_W(p^c), \tag{1}$$

which can be understood as the demand function because it is the percentage of the population who would subscribe at price p^c. Knowing the demand function $D(p^c)$, we can identify the price elasticity of demand, which is essential for pricing.[4] When X_i includes consumer characteristics like gender, we can infer the demand function from $F_W(W_i \mid X_i)$ in different market segments. For example, letting X_i be a categorical variable indicating gender,

$$\Pr(S_i = 1 \mid P = p^c, X_i = \text{Male}) = \Pr(W_i > p^c \mid X_i = \text{Male}) = 1 - F_W(p^c \mid X_i = \text{Male})$$

is the proportion of subscription among male consumers when price is p^c.

The key idea is that for a subscription-like service, the purchasing and usage (consumption) are separated in the sense that in data we can observe two subscribers who have different amount of usage, but paid the same price for the plan. As a result, though the price is constant for the subscribed plan, there is still variation in the price *per unit of usage*. This variation in price per unit of ex-post usage can either be across consumers or within a consumer across time periods.

Below we first express the WTP as a *known* function of the expected usage of the monthly gym membership if subscribed, and show how to identify the elasticities and the distribution of the WTP when the expected usage is observed. Then we discuss how to link the observed actual usage to the unobserved expected usage, so that the identification results still hold even when the expected usage is unobserved. In **??**, we include price variation, and show how price variation can help identify the functional form of the WTP as a function of the expected usage.

3.2 Expected usage is observed: cross section data case

Assumption 1 (Multiplicative form of the WTP)**.** *Let Q_i^* be the expected usage of the plan if subscribed. The WTP W_i has multiplicative form $W_i = \alpha_i Q_i^*$ with $\alpha_i > 0$, hence $\ln W_i = \ln \alpha_i + \ln Q_i^*$.*

To understand this assumption, let us talk about the underlying consumer decision process. Suppose consumer i needs to decide whether or not to get a monthly gym membership. First, depending on her distance to the gym, work schedule, she finds her expected number

[4] The price elasticity equals $D'(p^c)p^c/D(p^c)$, where $D'(p^c)$ is the first order derivative of $D(p^c)$.

of gym visits that is optimal for her, that is Q_i^*. Next, depending on her earnings, health awareness and gym facility, she decides her willingness to pay for an average gym visit, that is α_i. As a result, her willingness to pay for gym membership is her willingness to pay for one average visit, α_i, times her expected usage, Q_i^*. It should be emphasized that the multiplicative form $W_i = \alpha_i Q_i^*$ is not a specification of utility function of usage (hence implies constant marginal utility of usage). It's the outcome of consumer's decisions about usage and willingness to pay per unit of usage.

Consider the linear projection of $\ln \alpha_i$ on the vector of observable covariates X_{1i},

$$\ln \alpha_i = \beta' X_{1i} + U_i.$$

and U_i denotes the unobservable heterogeneity that affects one's WTP for one unit of usage. The first element of X_{1i} is unity with coefficient β_1, and $\mathrm{E}(U_i) = 0$. In the gym example, X_{1i} can include gender and some variables measuring the gym facility, and U_i can be unobserved health consciousness. When we do not observe any X_{1i}, $\alpha_i = e^{\beta_1 + U_i}$ is just a random coefficient of expected usage. When X_{1i} is "endogenous", that is X_{1i} and U_i are correlated, we need a vector of instrumental variables (IV) Z_i that is uncorrelated with U_i. By $S_i = \mathbb{I}(\ln W_i > \ln P)$, we have

$$S_i = \mathbb{I}(\beta' X_{1i} + \ln Q_i^* - \ln P + U_i > 0). \tag{2}$$

Before showing the formal results, consider a simple case to understand the intuition. Suppose **??** holds, $Q_i^* \perp\!\!\!\perp \alpha_i$, and Q_i^* is always observable. It suffices to know the distribution function $F_\alpha(a) \equiv \Pr(\alpha_i \leq a)$ of α_i in order to determine the CDF of $W_i = \alpha_i Q_i^*$.[5] We have

$$S_i = \mathbb{I}(\alpha_i Q_i^* > P) = \mathbb{I}(\alpha_i > P/Q_i^*),$$

hence

$$\Pr(S_i = 1 \mid Q_i^* = q) = \Pr(\alpha_i > P/q) = 1 - F_\alpha(P/q).$$

The object $\Pr(S_i = 1 \mid Q_i^* = q)$ is observable from data when Q_i^* is observable. So when Q_i^* has large enough variation, so that P/Q_i^* covers the support of α_i, we can identify F_α. Even when the variation of Q_i^* is limited, we can still identify F_α on a part of the support of α_i. Let S_i be the choice about whether or not to subscribe the monthly gym membership at price P, and let Q_i^* be the number of times one expects to visit the gym. Then P/Q_i^* is the "actual price" per visit for consumer i if she went to the gym only Q_i^* times per month.

[5] Since $Q_i^* \perp\!\!\!\perp \alpha_i$, the distribution of $\ln W_i = \ln \alpha_i + \ln Q_i^*$ is determined by the convolution of the observed distribution function of Q_i^* and the distribution function of $\ln \alpha_i$. Knowing the distribution function of $\ln W_i$, one knows the distribution function of W_i.

The intuition is that given the number of times i would visit being $Q_i^* = q$, the probability that WTP per visit α_i is greater than P/q (unknown object to be identified) must equal to the probability of $S_i = 1$ given $Q_i^* = q$ (an object observed in data). The above argument holds because a user subscribes if and only if she is willing to pay more than the "actual" price she ended paying. We need the independence condition $Q_i^* \perp\!\!\!\perp \alpha_i$, because otherwise we have only

$$\Pr(S_i = 1 \mid Q_i^* = q) = \Pr(\alpha_i > P/q \mid Q_i^* = q) = 1 - F_\alpha(P/q \mid Q_i^* = q).$$

So when Q_i^* is not independent of α_i, we can only identify the particular probabilities $F_\alpha(P/q \mid Q_i^* = q)$ for different values of Q_i^* rather than the entire distribution $F_\alpha(\cdot \mid Q_i^* = q)$. For example, we do not identify $F_\alpha(P/q' \mid Q_i^* = q) = \Pr(\alpha_i \leq P/q' \mid Q_i^* = q)$ for any $q' \neq q$.

We start with an identification result about the conditional CDF $F_W(W_i \mid X_{1i}, Z_i, Q_i^*)$ when the expected usage Q_i^* is observed for every individual in the sample. We will discuss how to link expected usage and actual observed usage later. Note that $F_W(W_i \mid X_{1i}) = \mathrm{E}(F_W(W_i \mid X_{1i}, Z_i, Q_i^*) \mid X_{1i})$ is also identified when we identify $F_W(W_i \mid X_{1i}, Z_i, Q_i^*)$ and observe Q_i^*. In the next subsection, we are going to deal with the unobserved expected usage and relax some of the strict conditions, but the main arguments and conclusion will be similar.

Assumption 2. *(i) (Observed expected usage). Q_i^* is observable;*

(ii) (Big support of consumption). The support of $\ln Q_i^* \mid (X_{1i}, Z_i)$ covers the support of $\ln(P/\alpha_i) \mid (X_{1i}, Z_i)$;

(iii) (Independence of consumption). $Q_i^* \perp\!\!\!\perp U_i \mid (X_{1i}, Z_i)$;

(iv) (Valid and relevant IV). $\mathrm{E}(Z_i U_i) = 0$, $\mathrm{E}(Z_i Z_i')$ is nonsingular, and $\operatorname{rank} \mathrm{E}(X_{1i} Z_i') = \dim(X_{1i})$.

(v) (Boundary conditions). The support of U_i contains 0, and $\lim_{u\to\infty} u[F_U(u \mid X_{1i}, Z_i) - 1] = \lim_{u\to-\infty} uF_U(u \mid X_{1i}, Z_i) = 0$.

The support condition **??** says that the variation of expected usage Q_i^* should be at least as large as the variation of the minimum amount of usage that consumer i thinks worth the price, because α_i can be interpreted WTP per unit of usage, i.e., the highest price one is willing to pay for one unit of usage. The conditional independence assumption **??** requires researchers to control enough consumers heterogeneity. **??** is a technical condition restricting the tail of the distribution of U_i.

Proposition 1 (Elasticities of the WTP: observed usage)**.** *Suppose* **??** *and* **??** *hold. We have*

$$\mathrm{E}(\ln W_i \mid X_{1i}) = \beta' X_{1i} + \mathrm{E}(\ln Q_i^* \mid X_{1i}) + \mathrm{E}(U_i \mid X_{1i}),$$

where $\mathrm{E}(U_i \mid X_{1i}) = \mathrm{E}(H_{1i} \mid X_{1i})$ *with*

$$H_{1i} = \frac{S_i - \mathbb{I}(\ln P - \beta' X_{1i} - \ln Q_i^* \leq 0)}{f_{\ln Q^*}(\ln Q_i^* \mid X_{1i}, Z_i)},$$

and $f_{\ln Q^*}(\ln Q_i^* \mid X_{1i}, Z_i)$ *is the probability density function (PDF) of* $\ln Q_i^*$ *given* (X_{1i}, Z_i)*.*

We have a 2-step least square (2SLS) formula for β*,*

$$\beta = [\mathrm{E}(\tilde{X}_{1i}\tilde{X}_{1i}')]^{-1}\,\mathrm{E}(\tilde{X}_{1i}Y_{1i}),$$

where $\tilde{X}_{1i}' = Z_i'[\mathrm{E}(Z_i Z_i')]^{-1}\,\mathrm{E}(Z_i X_{1i}')$*, and*

$$Y_{1i} = \frac{S_i - \mathbb{I}(\ln Q_i^* - \ln P > 0)}{f_{\ln Q^*}(\ln Q_i^* \mid X_{1i}, Z_i)}.$$

Note that β *is the probability limit of the 2SLS estimator of regressing* Y_{1i} *on* X_{1i} *using IV* Z_i*.*

When the expected usage Q_i^* is observed, this proposition states that we can estimate β by the following steps.

Step 1: Nonparametric estimation of $f_{\ln Q^*}(\ln Q_i^* | X_{1i}, Z_i)$*.* Obtain a nonparametric estimator $\hat{f}_{\ln Q^*}(\cdot \mid X_{1i}, Z_i)$ of the conditional PDF $f_{\ln Q^*}(\cdot \mid X_{1i}, Z_i)$ of $\ln Q_i^* \mid (X_{1i}, Z_i)$ from the sample (Q_i^*, X_{1i}, Z_i) for $i = 1, \ldots, n$.

Step 2: IV regression to estimate β*.* Estimate β by the 2SLS IV regression of $\hat{Y}_{1i}$ on X_{1i} using IV Z_i, where

$$\hat{Y}_{1i} = \frac{S_i - \mathbb{I}(\ln Q_i^* - \ln P > 0)}{\hat{f}_{\ln Q^*}(\ln Q_i^* \mid X_{1i}, Z_i)}.$$

?? provides a formula of $\mathrm{E}(\ln W_i | X_{1i})$, which can then be used to calculate the elasticities of the WTP to the change of X_{1i}. When X_{1i} is mean independent of U_i ($\mathrm{E}(U_i \mid X_{1i}) = 0$), and one approximates $\mathrm{E}(\ln Q_i^* \mid X_{1i})$ by a linear projection $\rho' X_{1i}$ (i.e. run a linear regression of $\ln Q_i^*$ on X_{1i}), we have $\mathrm{E}(\ln W_i \mid X_{1i}) = (\beta + \rho)' X_{1i}$, and the elasticities of the WTP to X_{1i} are $\beta + \rho$. In general, we can estimate $\mathrm{E}(\ln Q_i^* \mid X_{1i})$ and $\mathrm{E}(U_i \mid X_{1i}) = \mathrm{E}(H_{1i} \mid X_{1i})$ by nonparametric regression, and use their derivatives to calculate the elasticities.

Proposition 2 (CDF of the WTP: observed usage)**.** *Suppose* **??** *and* **??** *hold. Define the conditional choice probability (CCP) function,* $\pi_1(x_1, z, q) \equiv \mathrm{E}(S_i \mid X_{1i} = x_1, Z_i = z, \ln Q_i^* =$

q). *We have*

$$F_U(u \mid X_{1i}, Z_i) = 1 - \pi_1(X_{1i}, Z_i, \ln P - \beta' X_{1i} - u),$$
$$F_W(w \mid X_{1i}, Z_i, Q_i^*) = 1 - \pi_1(X_{1i}, Z_i, \ln P - \ln w + \ln Q_i^*),$$

and $F_W(w \mid X_{1i}, Z_i, Q_i^*) = F_U(\ln w - \beta' X_{1i} - \ln Q_i^* \mid X_{1i}, Z_i)$.

?? provides closed form formulas of $F_W(w \mid X_{1i}, Z_i, Q_i^*)$ in terms of the CCP $\pi_1(x_1, z, q)$, which can be estimated by a nonparametric regression of S_i on X_{1i}, Z_i and Q_i^*. **??** implies that the CCP $\pi_1(X_{1i}, Z_i, \ln Q_i^*)$ as non-decreasing in usage Q_i^*. To see this, as a CDF, $F_W(w \mid X_{1i}, Z_i, Q_i^*)$ is non-decreasing in w. By the formula of $F_W(w \mid X_{1i}, Z_i, Q_i^*)$ in terms of CCP, we can conclude that π_1 is non-decreasing in usage for any given (X_{1i}, Z_i). Because we can always estimate the CCP function, this property can be used to check our model specification. If we find that $\pi_1(X_{1i}, Z_i, \ln Q_i^*)$ is decreasing in usage Q_i^* for certain intervals of Q_i^* given (X_{1i}, Z_i), that would suggest that the model specification might be questionable (e.g., we have not controlled for enough consumer characteristics X_{1i} or unobservable factors play a big role).

Remark 1 (What if there is no variation of consumption?). When every individual consumes the same amount, e.g., purchasing and consumption are not separated, for each purchase, Q_i^* will be a constant, denoted by q^* here. In this case, there is no variation in consumption or in price. **??** and **??** fail because the *support condition does not hold*. The consequence is that we cannot identify the distribution of the WTP. To see this, let $\ln \alpha_i = U_i$ for simplicity of exposition. We have

$$S_i = \mathbb{I}(\ln q^* - \ln P + U_i > 0).$$

We then have $F_U(\ln P - \ln q^*) = 1 - \mathrm{E}(S_i)$, but that is all we can identify. We only identify the CDF of F_U *evaluated at one point*, $\ln P - \ln q^*$. That is we can only identify $F_W(p \,|\, Q_i^* = q^*) = 1 - \mathrm{E}(S_i)$. □

3.3 Expected usage is observed: panel data case

For the rest of the paper, assume that we observe panel data about consumers subscription decisions. Let W_{it} be consumer i's WTP for the subscription service in billing period t, and let Q_{it}^* be her expected usage of the service in the billing period t. The other notation such as α_{it}, X_{i1t}, and U_{it} are defined similarly. We still maintain the assumption that the expected usage Q_{it}^* is observed in this subsection.

There could be two sources of the unobserved heterogeneity U_{it} in the valuation about the subscribed service. One is time invariant consumer specific unobserved heterogeneity,

and the other is the unobserved product characteristic that is common to all consumers. We let ω_i and ξ_t to denote the former and the latter, respectively, and let $U_{it} = \omega_i + \xi_t$. Below, we are going to show that with panel data we indeed can separately identify the distributions of ω_i and ξ_t under certain conditions. To simplify the exposition, we omit X_{i1t} and assume $Q^*_{it} \perp\!\!\!\perp U_{it}$ below.

The idea is to note that ω_i is time invariant in the above specification. Below, we first identify the values of the difference $\xi_{t+1} - \xi_t$ by the time variation (**??** below). If one is willing to assume an initial value ξ_0, then one identifies each value of ξ_t. Then the distribution of ω_i is easily obtained from by distribution of $U_{it} - \xi_t$ since we have identified the distribution of U_i. As an alternative, one can assume that ξ_t is serially independent and identically distributed, we then can identify its distribution from this difference by using the constrained deconvolution (**?**). The constraint is that $F(\xi_t) = F(\xi_{t+1})$. After identifying the distribution $F(\xi_t)$, we can identify the distribution ω_i by deconvolution of the distribution of $U_{it} = \omega_i + \xi_t$ from assuming that $\xi_t \perp\!\!\!\perp \omega_i$.

Below, we provide a formula of $\xi_{t+1} - \xi_t$. Based on an observation made by **?**, page 147, we show that

$$\lambda_{it} = U_{it}, \tag{3}$$

where

$$\lambda_{it} = \int_{-\infty}^{\infty} \mathbb{I}(\ln Q^*_{it} - \ln P > -U_{it}) - \mathbb{I}(\ln Q^*_{it} - \ln P > 0)\ \mathrm{d}(\ln Q^*_{it}).$$

The proof of **??** is in the appendix. Because $U_{it} = \omega_i + \xi_t$, we then have

$$\lambda_{i,t+1} - \lambda_{it} = \xi_{t+1} - \xi_t.$$

Note that taking integral with respect ω_i both sides of the equation, we have

$$\int_\omega \lambda_{i,t+1}\ \mathrm{d}\,F(\omega) - \int_\omega \lambda_{it}\ \mathrm{d}\,F(\omega) = \xi_{t+1} - \xi_t,$$

because $\xi_t = \int_\omega \xi_t\ \mathrm{d}\,F(\omega)$. Next, we show that

$$\begin{aligned}
\int_\omega \lambda_{it}\ \mathrm{d}\,F(\omega) &= \int_\omega \int_{-\infty}^{\infty} \mathbb{I}(\ln Q^*_{it} - \ln P + \xi_t > -\omega_i) - \mathbb{I}(\ln Q^*_{it} - \ln P > 0)\ \mathrm{d}(\ln Q^*_{it})\ \mathrm{d}\,F(\omega) \\
&= \int_\omega \int_{-\infty}^{\infty} \frac{S_{it} - \mathbb{I}(\ln Q^*_{it} - \ln P > 0)}{f_{\ln Q,t}(Q^*_{it})} f_{\ln Q,t}(\ln Q^*_{it})\ \mathrm{d}(\ln Q^*_{it})\ \mathrm{d}\,F(\omega) \\
&= \mathrm{E}_t\left(\frac{S_{it} - \mathbb{I}(\ln Q^*_{it} - \ln P > 0)}{f_{\ln Q,t}(\ln Q^*_{it})}\right) \\
&\equiv \mathrm{E}_t(\tilde{Y}_{1it}).
\end{aligned}$$

Here $f_{\ln Q,t}(Q^*_{it})$ is the density function of Q^*_{it} in period t. The expectation $\mathrm{E}_t(\cdot)$ is taken with respect to the distribution of (S_{it}, Q^*_{it}) in period t only, because ξ_t is held as constant. We hence have the conclusion that

$$\xi_{t+1} - \xi_t = \mathrm{E}_{t+1}(\tilde{Y}_{1i,t+1}) - \mathrm{E}_t(\tilde{Y}_{1it}). \tag{4}$$

Obviously, the right-hand-side (RHS) is estimable from data by sample averages.

3.4 Link observed usage and expected usage: past usage is always observable

In the study of churn decisions ($S_{it} = 0$ if i churn before the beginning of period t, and $S_{it} = 1$ otherwise), the sample contains the historical usages of all consumers. Let Q_{it} be the observed consumption in billing period t.

Assumption 3 (Expected usage and observed usage: I)**.** *Assume* $\ln Q^*_{it} = \ln Q_{i,t-1} + \zeta_{it}$, *and* $\zeta_{it} \perp\!\!\!\perp Q_{i,t-1}$.[6]

?? implies that the subscription choice can be written in terms of the actual usage as follows,

$$S_{it} = \mathbb{I}(\beta' X_{i1t} + \ln Q_{i,t-1} - \ln P + (U_{it} + \zeta_{it}) > 0). \tag{5}$$

Viewing $Q_{i,t-1}$ and $\tilde{U}_{it} \equiv U_{it} + \zeta_{it}$ in the above display as Q^*_{it} and U_{it} of **??**, **??** and **??** can still be applied. Hence the elasticities and the distribution function of the WTP can be identified.

3.5 Link observed usage and expected usage: usage is only observable for subscribers

In many applications, we do not observe the usage by non-subscribers at all. Below, we assume that although we cannot observe the usage by each individual, we observe consumer and product characteristics X_{2it} for everyone. Using the data about subscribers whose usage are observed, we project the usage onto the space of X_{2it}, then impute the "usage" driven by the variation of X_{2it} for each individual. Sample selection (subscribers and non-subscribers are different in unobserved characteristics that affect usage) is not a hard issue, when we have panel data about usage and characteristics X_{2it}. In the gym example, X_{2it} can include the distance to the gym, leisure time and weather.

[6] A special case of **??** is that $\zeta_{it} = 0$, and consumers determine their expected based on the usage in the previous month.

Assumption 4 (Expected usage and observed usage: II). *(i) Assume that the log of expected consumption* $\ln Q_{it}^*$ *has the following reduced form,*

$$\ln Q_{it}^* = \gamma' X_{2it} + V_i,$$

where the unobserved fixed effect V_i *can be correlated with* U_{it} *in the specification of the WTP for one unit of consumption* (α_i) *to allow for selection. Assume that* $\mathrm{E}(V_i) = 0$.

(ii) Let Q_{it} *be the observed consumption when* $S_{it} = 1$*, and assume*

$$\ln Q_{it} = \ln Q_{it}^* + \varepsilon_{it},$$

where ε_{it} *is serially uncorrelated random shock.*

(iii) Assume strict exogeneity $\mathrm{E}(\varepsilon_{it} \mid X_i, U_i, V_i) = 0$*, where* $X_i = (X_{i1}', \ldots, X_{iT}')'$ *and* $U_i = (U_{i1}, \ldots, U_{iT})'$*. Here* X_{it} *is the union of* X_{i1t} *and* X_{2it}*.*

(iv) All covariates X_{2it} *are time varying, and* X_{2it} *are observable for both subscribers and non-subscribers.*

?? is of course restrictive (e.g., by assuming all elements of X_{2it} are time varying, X_{2it} does not even include intercept term). In **??**, one will see that it is straightforward to extend the analysis to allow for time invariant covariates.

Let $S_i = (S_{i1}, \ldots, S_{iT})'$ be the vector of subscription choice in the T sampling periods. Because S_i is a function of X_i, U_i, V_i, we have

$$\mathrm{E}(\varepsilon_{it} \mid X_i, V_i, S_i) = \mathrm{E}(\mathrm{E}(\varepsilon_{it} \mid X_i, U_i, V_i) \mid X_i, V_i, S_i) = 0.$$

Let $\overline{\ln Q}_i = T^{-1} \sum_{t=1}^T \ln Q_{it}$ and $\bar{X}_{2i} = T^{-1} \sum_{t=1}^T X_{2it}$. Using $\varepsilon_{it} = \ln Q_{it} - \gamma' X_{2it} - V_i$, we have $\mathrm{E}(\ln Q_{it} - \gamma' X_{2it} - V_i \mid X_i, V_i, S_i = 1) = 0$, and

$$\mathrm{E}((\ln Q_{it} - \overline{\ln Q}_i) - \gamma'(X_{2it} - \bar{X}_{2i}) \mid X_i, V_i, S_i = 1) = 0,$$

from which the γ associated with time varying X_{2it} can be identified and estimated by the usual fixed effect (FE) estimator using the selected sample ($S_{it} = 1$). Of course FE estimator cannot identify the γ associated with time invariant X_{2it}.

Let

$$\tilde{Q}_{it} \equiv \gamma' X_{2it} \qquad \text{and} \qquad \eta_{it} \equiv U_{it} + V_i.$$

We have $\ln Q_{it}^* = \tilde{Q}_{it} + V_i$, and

$$S_{it} = \mathbb{I}(\beta' X_{i1t} + \tilde{Q}_{it} - \ln P + \eta_{it} > 0).$$

Because $\tilde{Q}_{it}$ is identified for subscribers and non-subscribers, we can view $\tilde{Q}_{it}$ as a special regressor and apply similar arguments in the last section to the above display. Note that $\tilde{Q}_{it}$ should be only understood as the log of the identified consumption component that is driven only by X_{2it}.

Assumption 5. *(i) (Big support of consumption). The support of* $\tilde{Q}_{it} \mid (X_{i1t}, Z_{it})$ *covers the support of* $\ln(P/\alpha_{it}) - V_i \mid (X_{i1t}, Z_{it})$*;*

(ii) (Independence of consumption). $X_{2it} \perp\!\!\!\perp (U_{it}, V_i) \mid (X_{i1t}, Z_{it})$*;*

(iii) (Valid and relevant IV). $\mathrm{E}(Z_{it}\eta_{it}) = 0$*,* $\mathrm{E}(Z_{it}Z_{it}')$ *is nonsingular, and* $\operatorname{rank} \mathrm{E}(X_{i1t}Z_{it}') = \dim(X_{i1t})$*.*

(iv) (Boundary conditions). The support of η_{it} *contains* 0*, and* $\lim_{\eta\to\infty} \eta[F_\eta(\eta \mid X_{i1t}, Z_i) - 1] = \lim_{\eta\to-\infty} \eta F_\eta(\eta \mid X_{i1t}, Z_i) = 0$*.*

?? and **??** imply that X_{2it} must have excluded variables that affect usage but do not affect the valuation of one unit of usage (hence does not belong to X_{i1t}), and such excluded variables are independent of U_{it} and V_i given X_{i1t} and Z_{it}. If X_{2it} is a subset of X_{i1t}, the support condition of **??** fails as the support of $\tilde{Q}_{it} = \gamma' X_{2it} \mid (X_{i1t}, Z_{it})$ includes only one point. In gym usage example, such an excluded variable can be weather in the residence of consumer i. In music streaming example, such excluded variables could be the measurement about the access to internet (e.g., WiFi or 4G network coverage, quota of cellular data). The access to internet affects the usage but do not affect one's valuation of one unit of usage, say one song, after controlling variables like income.

Proposition 3 (Elasticities of the WTP: unobserved usage)**.** *Suppose* **??**, **??** *and* **??** *hold. We have*

$$\mathrm{E}(\ln W_{it} \mid X_{i1t}, X_{2it}) = \beta' X_{i1t} + \gamma' X_{2it} + \mathrm{E}(\eta_{it} \mid X_{i1t}, X_{2it}),$$

where $\mathrm{E}(\eta_{it} \mid X_{i1t}, X_{2it}) = \mathrm{E}(H_{2it} \mid X_{i1t}, X_{2it})$ *with*

$$H_{2it} = \mathrm{E}\left(\frac{S_{it} - \mathbb{I}(\ln P - \beta' X_{i1t} - \tilde{Q}_{it} \le 0)}{f_{\tilde{Q}}(\tilde{Q}_{it} \mid X_{i1t}, Z_{it})} \,\middle|\, X_{i1t}, Z_{it} \right).$$

Here $f_{\tilde{Q}}(\tilde{Q}_{it} \mid X_{i1t}, Z_{it})$ *is the PDF of* $\tilde{Q}_{it}$ *given* (X_{i1t}, Z_{it})*.*

We have the following 2SLS formula for β*,*

$$\beta = [\mathrm{E}(\tilde{X}_{1it}\tilde{X}_{1it}')]^{-1} \mathrm{E}(\tilde{X}_{1it} Y_{2it}),$$

where

$$Y_{2it} = \frac{S_{it} - \mathbb{I}(\tilde{Q}_{it} - \ln P > 0)}{f_{\tilde{Q}}(\tilde{Q}_{it} \mid X_{i1t}, Z_{it})}.$$

Again, β *is the probability limit of the 2SLS estimator of regressing* Y_{2it} *on* X_{1it} *using IV* Z_{it}*.*

Similar to **??**, we now have a formula of $\mathrm{E}(\ln W_{it}|X_{i1t}, X_{2it})$, which can be used to calculate the elasticities of the WTP to X_{i1t} and X_{2it}. When X_{i1t} and X_{2it} are mean independent of η_{it}, the elasticities of the WTP depends only on β and γ. Using the usage data, one can estimate γ and β is can be estimated by a 2SLS regression. The following steps outline the procedure.

Step 1: Fixed effect estimator for log-usage. Using the data of the subscribers, estimate the following panel data model

$$\ln Q_{it} = \gamma' X_{2it} + V_i + \varepsilon_{it},$$

with fixed effect estimator. Let $\hat{\gamma}$ be the estimator of γ, and let $\hat{Q}_{it} = \hat{\gamma}' X_{2it}$ for all i (subscribers and non-subscribers). When X_{2it} includes the unity term and other time invariant covariates, the associated parameters are not identified. We can apply the arguments of **??**.

Step 2: Nonparametric estimation of $f_{\tilde{Q}}(\tilde{Q}_{it} \mid X_{i1t}, Z_{it})$. Obtain a nonparametric estimator $\hat{f}_{\tilde{Q}}(\cdot \mid X_{i1t}, Z_{it})$ of the conditional PDF $f_{\tilde{Q}}(\cdot \mid X_{i1t}, Z_{it})$ of $\tilde{Q}_{it} \mid (X_{i1t}, Z_{it})$ from the sample $(\hat{Q}_{it}, X_{i1t}, Z_{it})$ for $i = 1, \ldots, n$ and $t = 1, \ldots, T$.

Step 3: IV regression to estimate β. Estimate β by the 2SLS IV regression of $\hat{Y}_{2it}$ on X_{i1t} using IV Z_{it}, where

$$\hat{Y}_{2it} = \frac{S_{it} - \mathbb{I}(\hat{Q}_{it} - \ln P > 0)}{\hat{f}_{\tilde{Q}}(\hat{Q}_{it} \mid X_{i1t}, Z_{it})}.$$

Proposition 4 (CDF of the WTP: unobserved usage)**.** *Suppose* **??**–**??** *hold. Define the CCP function, $\pi_2(X_{i1t}, Z_{it}, \tilde{Q}_{it}) = \mathrm{E}(S_{it} \mid X_{i1t}, Z_{it}, \tilde{Q}_{it})$. We have*

$$F_\eta(\eta \mid X_{i1t}, Z_{it}) = 1 - \pi_2(X_{i1t}, Z_{it}, \ln P - \beta' X_{i1t} - \eta),$$
$$F_W(w \mid X_{it}, Z_{it}) = 1 - \pi_2(X_{i1t}, Z_{it}, \ln P - \ln w + \gamma' X_{2it}),$$

and $F_W(w \mid X_{it}, Z_{it}) = F_\eta(\ln w - \beta' X_{i1t} - \gamma' X_{2it} \mid X_{i1t}, Z_{it})$.

Remark 2 (What are really X_{i1t} and X_{2it}?). We now show how to relax **??** and **??** by allowing X_{2it} to include time invariant covariates and/or to include covariates that are correlated with either U_{it} or V_i. The most obvious example is that X_{2it} includes the intercept (unity) term. Suppose

$$\ln Q^*_{it} = \gamma'_a X_{2it,a} + \gamma'_b X_{2it,b} + V_i,$$

where $X_{2it,b}$ is the vector of time varying covariates, and the vector $X_{2it,a}$ includes time invariant covariates (gender, location) or the covariates that are not independent of (U_{it}, V_i) given (X_{i1t}, Z_{it}). By the fixed effect estimator, we can still identify γ_b, but not necessarily γ_a. Under the above specification, we have

$$S_{it} = \mathbb{I}((\beta' X_{i1t} + \gamma'_a X_{2it,a}) + \gamma'_b X_{2it,b} - \ln P + \eta_{it} > 0). \tag{6}$$

Observe that it is similar to the model

$$S_{it} = \mathbb{I}(\beta' X_{i1t} + \gamma' X_{2it} - \ln P + \eta_i > 0). \tag{7}$$

We can redefine β, X_{i1t} and X_{2it} in **??** so that $\beta' X_{i1t} + \gamma_a' X_{2it,a}$ of **??** becomes $\beta' X_{i1t}$ of **??**, and $\gamma_b' X_{2it,b}$ of **??** becomes $\gamma' X_{2it}$ of **??**, that is also $\tilde{Q}_{it}$.

When the support of $\gamma_b' X_{2it,b} | (X_{i1t}, X_{2it,a}, Z_{it})$ covers the support $(\ln P - \beta' X_{i1t} - \gamma_a' X_{2it,a} - \eta_{it}) \mid (X_{i1t}, X_{2it,a}, Z_{it})$, $X_{2it,b} \perp\!\!\!\perp (U_{it}, V_i) \mid (X_{i1t}, X_{2it,a}, Z_{it})$, and there are enough number of IVs, we can apply **??** and **??** to identify $F_W(w \mid X_{it}, Z_{it})$ and $\mathrm{E}(\ln W_{it} \mid X_{it}, Z_{it})$.

The caveat is that if X_{i1t} and $X_{2it,a}$ has intersections (e.g., when both X_{i1t} and $X_{2it,a}$ have unity term), and γ_a cannot be identified from consumption data (e.g. due to time invariant nature of such variables), we cannot separately identify β and γ_a from the subscription choice. Instead, we can only identify $\beta + \gamma_a$. □

3.6 Discussion about applications

3.6.1 Pricing a shorter subscription service based on higher frequency usage data

Usually consumption has seasonality and serial correlation. When the distribution of consumption of a subscription plan varies within its billing period (e.g., yearly plan), consumers may want a plan of shorter length (e.g., monthly plan) or a plan for a particular period (holiday season). With the access to high frequency consumption data (e.g., daily consumption data), we show that it is possible to identify the distribution of the WTP for plans of varying lengths or plans for certain specific periods.

A reasonable question here is why we need a model to set pricing for different lengths of plan. For example, why not use a simple approach of just dividing the price by the period length, for example a monthly subscription should be priced at 1/12 of an annual subscription, since the mean monthly usage is on average 1/12 of annual usage? This argument does not hold, since we need the *distribution* of monthly usage to be exactly the distribution of 1/12 of the annual usage for this to work. When there is serial correlation in usage across different periods, this will not hold. For example, suppose the 12 monthly usage variables have identical distributions, the variance of 1/12 of the annual usage is always smaller than the variance of monthly usage unless the 12 monthly usages are perfectly correlated. Thus, in general the optimal price for a monthly plan may be different from that of scaled down annual price.

Another reason for varied valuation over time periods is that peak and off-peak usage might be different. To fix the idea, suppose the subscription period is one year, and it

consists of peak and off-peak periods. Consumers use the plan more during the peak period. Also, assume that X_{it} are exogenous, so $Z_{it} = X_{it}$. Let t denote one year, and let t_p and t_{op} denote the peak and off-peak periods in year t. Similarly, let Q_{it_p} and $Q_{it_{op}}$ be the observed consumption in the two periods, and $Q_{it} = Q_{it_p} + Q_{it_{op}}$.

From the actual yearly consumption Q_{it}, we can first identify the distribution $F_\eta(\eta_{it}|X_{i1t})$. We then can identify the distribution function of the WTP for plan during peak and off-peak period. Taking the peak period for example, the log of the WTP for the plan of the peak period is

$$\ln W_{it_p} = \ln \alpha_{it_p} + \tilde{Q}_{it_p} + V_i = \beta' X_{1it_p} + \gamma_p' X_{2it_p} + V_i + U_{it_p},$$

where γ_p can be identified from the consumption in peak period. If we assume that $U_{it_p} = U_{it}$, we have $V_i + U_{it_p}$ and η_{it} have identical distribution function $F_\eta(\eta|X_{i1t})$. We then can identify the distribution function of W_{it_p}:

$$F_{W_{t_p}}(w \mid X_{i1t} = x_1, X_{2it} = x_2) = F_\eta(\ln w - \beta' x_1 - \gamma_p' x_2 \mid X_{i1t} = x_1).$$

Because the assumption $U_{it_p} = U_{it}$ is important, it is instructive to provide examples, in which this assumption holds and fails. The interpretation of U_{it} or U_{it_p} depends on what have been controlled by X_{i1t} or X_{1it_p}. In the gym subscription example, first consider the following specification

$$\ln \alpha_{it} = \beta_1 + \beta_2 GymAval_{it} + \beta_3 INC_{it}, \quad \text{and} \quad \ln \alpha_{it_p} = \beta_1 + \beta_2 GymAval_{it_p} + \beta_3 INC_{it},$$

where INC_{it} is consumer i's annual income in period t, $GymAval_{it}$ is a measurement of fitness equipments availability during period t, and $GymAval_{it_p}$ is the availability during peak period t_p. If we observe $GymAval_{it}$ and $GymAval_{it_p}$, but do not observe INC_{it}, $X_{i1t} = GymAval_{it}$ and $U_{it} = \beta_3 INC_{it} = U_{it_p}$. On the other hand, if we observe INC_{it}, but do not observe $GymAval_{it}$ or $GymAval_{it_p}$, $X_{i1t} = INC_{it}$, $U_{it} = \beta_2 GymAval_{it}$, and $U_{it_p} = \beta_2 GymAval_{it_p}$. It is then unreasonable to claim that $U_{it} = U_{it_p}$ because fitness gears availability is lower than usual.

3.6.2 Evaluate the money-metric effect of product changes

We have shown how to identify the demand function $D(p)$ in **??**. Demand function can be used to estimate the money-metric effect of product change, which will be useful given that product improvement is usually costly. Suppose a music streaming company wants to evaluate the effects of streaming music at high quality (HQ) which requires more bandwidth on the WTP. For simplicity, suppose $X_{it} = (X_{i1t}, X_{2it})'$, and X_{i1t} is a dummy variable that equals 1 if the music was streamed at HQ and 0 otherwise, and X_{2it} is an excluded variable

with large support. The company can run experiments in two cities with $X_{i1t} = 0$ (non-HQ) and 1 seconds (HQ). Our method can identify $F_W(W_{it} \mid X_{i1t} = 1)$ and $F_W(W_{it} \mid X_{i1t} = 0)$.[7] Then we have demand functions for the music streaming service with and without HD, $D(P \mid X_{i1t} = 1) = 1 - F_W(P \mid X_{i1t} = 1)$ and $D(P \mid X_{i1t} = 0) = 1 - F_W(P \mid X_{i1t} = 0)$. Then $(D(P \mid X_{i1t} = 1) - D(P \mid X_{i1t} = 0))/D(P \mid X_{i1t} = 0)$ is the percentage change in the revenue when the music is streamed at HQ.

3.6.3 Pricing based on usage

Using individual usage data, we have shown that we can identify $F_W(W_{it} \mid X_{1it}, Q^*_{it})$, hence $F_W(W_{it} \mid Q^*_{it})$. Knowing consumer's WTP given their usage, we can specify a pricing model based on usage and derive the corresponding demand curve and marginal revenue.

For simplicity, assume that there is no X_{i1t}, the subscription choice follows $S_{it} = \mathbb{I}(\ln Q^*_{it} - \ln P + U_{it} > 0)$, and $Q^*_{it} \perp\!\!\!\perp U_{it}$. We have identified $F_U(u)$. Consider the following counterfactual usage based pricing model. At price P_1, one can use the subscription service up to Q^h amount. With a higher price P_2, one has unlimited usage of the service. Assume that the distribution of the usage is unchanged in this counterfactual pricing model.[8]

Under this pricing model, the demand consists of the demand for the plan (called plan 1) with quota Q^h and price P_1 and the plan (called plan 2) without limit and price P_2. The demand for plan 1 is proportional to

$$\int_{-\infty}^{Q^h} P_A f_{\ln Q^*}(q)\ \mathrm{d}\, q + \int_{Q^h}^{Q^h + \ln(P_2/P_1)} P_B f_{\ln Q^*}(q)\ \mathrm{d}\, q,$$

where

$$\begin{aligned} P_A &\equiv \Pr(q - \ln P_1 + U_{it} > 0) = 1 - F_U(\ln P_1 - q), \\ P_B &\equiv \Pr(Q^h - \ln P_1 + U_{it} > 0) = 1 - F_U(\ln P_2 - Q^h). \end{aligned}$$

The demand for plan 2 is proportional to

$$\int_{Q^h + \ln(P_2/P_1)}^{\infty} P_C f_{\ln Q^*}(q)\ \mathrm{d}\, q,$$

where

$$P_C \equiv \Pr(q - \ln P_2 + U_{it} > 0) = 1 - F_U(\ln P_2 - q).$$

[7] We first identify $F_W(w \mid X_{i1t}, X_{2it})$, then we use $F_W(w \mid X_{i1t}) = \mathrm{E}(F_W(w \mid X_{i1t}, X_{2it}) \mid X_{i1t})$.

[8] It can be problematical to make the assumption that the distribution of consumption does not change when consumers face the above counterfactual usage based pricing model. **?** have found consumers will adjust their consumption to adapt the threshold imposed by the three-part tariff pricing model. One has to specify a full usage model to address this concern.

Knowing the demand function, it is straightforward to derive the marginal revenue and find the optimal (P_1, P_2, Q^h) when the marginal cost is known.

4 What usage tracking data cannot tell us but price variation can

All of our previous results rely on the multiplicative form of the WTP $W_{it} = \alpha_{it} Q^*_{it}$, or equivalently the additive form $\ln W_{it} = \ln \alpha_{it} + \ln Q^*_{it}$. One natural question is that can we allow for nonlinear relationship between $\ln W_{it}$ and $\ln Q^*_{it}$? The answer of this question depends on whether or not we have price variation at all in data. First, we are going to argue that without any price variation, subscription/churn choices and usage tracking data cannot tell us the functional form of the log of WTP $\ln W_{it}$ as a function of the log of usage $\ln Q^*_{it}$. Moreover, without knowing the functional form, the distribution of the WTP cannot be identified. Secondly, we show how to use price variation to parametrically identify the functional form of $\ln W_{it}$ as a function $\ln Q^*_{it}$. We end up this section with the identification of the outside option valuation as an application of the general result.

4.1 What usage tracking data cannot tell us

We focus on the case where we observe the expected usage for all sampled individuals. Suppose one lets

$$\ln W_{it} = g(\ln \alpha_{it} + \ln Q^*_{it}) = g(\beta' X_{i1t} + U_{it} + \ln Q^*_{it}),$$

where g is some strictly increasing differentiable function. The multiplicative form in **??** assumes that g is an identity function. As a motivation for general functional form g, consider the Taylor expansion of $g(\ln \alpha_{it} + \ln Q^*_{it})$ at $Q^*_{it} = 0$. We have

$$\ln W_{it} = g(\ln \alpha_{it}) + g'(\ln \alpha_{it}) \ln Q^*_{it} + g''(\ln \alpha_{it})/2(\ln Q^*_{it})^2 + \ldots,$$

which gives us familiar concave utility function of consumption.

Below we show that without knowing g, one cannot identify the distribution of the WTP W_{it}. To see this, let's ignore X_{i1t} and let $\ln \alpha_{it} = U_{it}$. We have

$$S_{it} = \mathbb{I}(\ln W_{it} > \ln P) = \mathbb{I}(g(U_{it} + \ln Q^*_{it}) > \ln P) = \mathbb{I}(U_{it} + \ln Q^*_{it} > g^{-1}(\ln P)).$$

We then can still apply **??**, and obtain the CDF of U_{it} as follows,

$$F_U(u; g) = 1 - \mathrm{E}(S_{it} \mid \ln Q^*_{it} = g^{-1}(\ln P) - u).$$

The conditional CDF of the WTP W_{it} given Q^*_{it} is

$$F_W(w \mid Q^*_{it} = q; g) = 1 - \mathrm{E}(S_{it} \mid \ln Q^*_{it} = g^{-1}(\ln P) - g^{-1}(\ln w) + \ln q).$$

So the identified $F_W(\cdot \mid Q^*_{it}; g)$ will also change with respect to g. By the law of iterated expectation, the marginal CDF of the WTP W_{it} is as follows,

$$F_W(w; g) = \int_{-\infty}^{\infty} \Big[1 - \mathrm{E}(S_{it} \mid \ln Q^*_{it} = g^{-1}(\ln P) - g^{-1}(\ln w) + \ln q)\Big] f_{\ln Q^*}(\ln q)\ \mathrm{d}(\ln q).$$

So the identified $F_W(w; g)$ will also change with respect to the specification of g.

4.2 Price variation across markets

We have seen that certain tight specification, like multiplicative form in **??**, is necessary to identify the distribution of the WTP. We are going to argue that the price variation can help to relax this specification, and the extent of the relaxation depends on the observed variation of price.

When we observe data from multiple markets, it is possible that the same subscription service has different price. For example, the PlayStation Plus (Sony's prime service for PlayStation players) costs \$59.99 annually plus tax in the US and £49.99 annually in the UK. There could also be temporal price variation for the same market. For example, PlayStation Plus in the UK was priced at £39.99 before September 2017.

Suppose in the sample, there are M markets across which the price varies. Let P_m be the price in market m, and the other notation including $W_{it,m}$, $X_{i1t,m}$, $X_{i2t,m}$, and $\tilde{Q}_{it,m}$ are defined similarly for each market m. Assume that the log of WTP for the service in market $m = 1, \ldots, M$ equals,

$$W_{it,m} = g(\beta' X_{1it,m} + \tilde{Q}_{it,m} + U_{it,m} + V_{i,m}; \delta) = g(\beta' X_{1it,m} + \tilde{Q}_{it,m} + \eta_{it,m}; \delta).$$

Let $g(\cdot; \delta)$ be a strictly increasing function known up to a finitely dimensional vector of parameters δ. Assume $\tilde{Q}_{it,m}$ has been identified. The subscription decision $S_{it,m}$ will depend on the local price P_m in the market m where consumer i resides. We then have

$$S_{it,m} = \mathbb{I}(\beta' X_{1it,m} + \tilde{Q}_{it,m} + \eta_{it,m} > g^{-1}(P_m; \delta)). \tag{8}$$

Note that δ can include the valuation of the outside option, which could even be market specific when there is enough price variation within a market. As we will show in the next subsection that when the multiplicative form of WTP in **??** holds, we only need two distinct prices in one market to identify the valuation of the outside option in that market. So if the

multiplicative form assumption holds and each market has more than two different prices, we can identify the valuation of the outside option for each market. Our previous results were derived essentially by letting $g(c;\delta) = e^c$. For simplicity, assume $X_{1it,m}$ is exogenous, so we ignore Z_{it} below. By the same arguments in the previous sections, we have

$$\beta = \mathrm{E}(X_{1it,m}X'_{1it,m})^{-1}\,\mathrm{E}(X_{1it,m}Y_{2it,m}),$$

where

$$Y_{2it,m} = \frac{S_{it,m} - \mathbb{I}(\tilde{Q}_{it,m} - g^{-1}(P_m;\delta) > 0)}{f_{\tilde{Q}_m}(\tilde{Q}_{it,m} \mid X_{1it,m})}.$$

For simplicity, suppose $X_{1it,m}$ has the same distribution across different markets, hence $\mathrm{E}(X_{1it,m}X'_{1it,m})$ is identical across m. We then have a set of moment equations of δ,

$$\mathrm{E}(X_{1it,j}Y_{2it,j}) - \mathrm{E}(X_{1it,k}Y_{2it,k}) = 0, \qquad 1 \le j < k \le M, \tag{9}$$

for any pair of two markets (j,k). The number of moment equations depends on the dimension of X_{i1t} and the number of markets. When there are enough number of markets with different prices, we can identify the parameters δ of $g(\cdot;\delta)$. Once g is identified, the identification of the distribution of W_{it} follows from the earlier results.

4.3 Identify the utility of the outside option by price variation

As one application of the above arguments about the role of price variation, we show how to identify the WTP for the outside option, μ, by price variation within the same markets. In some applications, we can observe multiple prices for the same subscription service for various reasons. For example, in our music streaming empirical application, there are three common prices 99, 129 and 149 New Taiwan dollars (NT$) per month, among which 149 is the usual price, 129 is the price for customers of certain cellular carriers, and new users have special price 99 for the first two months.

Suppose there are $P_1, \ldots, P_M$ prices of the same subscription in a market. Consumers facing price P_m make the subscription decision based on the following rule,

$$S_{it,m} = \mathbb{I}(\beta' X_{1it,m} + \tilde{Q}_{it,m} + \eta_{it,m} > \ln(P_m + \mu)).$$

Here the outside option valuation μ is unknown, but invariant of the prices. This specification is a special case of **??** by letting $g(c;\delta) = e^c - \delta$ and $\delta = \mu$. We then can use the moment **??** to identify μ. To be concrete, suppose $X_{1it,m} = 1$ only. **??** reads

$$\mathrm{E}\left(\frac{S_{it,j}}{f_{\tilde{Q}_j}(\tilde{Q}_{it,j})} - \frac{S_{it,k}}{f_{\tilde{Q}_k}(\tilde{Q}_{it,k})}\right) = \mathrm{E}\left(\frac{\mathbb{I}(\tilde{Q}_{it,j} - \ln(P_j + \mu) > 0)}{f_{\tilde{Q}_j}(\tilde{Q}_{it,j})} - \frac{\mathbb{I}(\tilde{Q}_{it,k} - \ln(P_k + \mu) > 0)}{f_{\tilde{Q}_k}(\tilde{Q}_{it,k})}\right), \tag{10}$$

for any $1 \leq j < k \leq M$.

To see more intuition, it can be shown that

$$\text{RHS of ??} = \ln\left(\frac{P_k + \mu}{P_j + \mu}\right),$$

and

$$\text{LHS of ??} = \int_{\mathcal{Q}} \int_{\eta} (S_{it,j} - S_{it,k}) \, \mathrm{d}\, F_{\eta}(\eta) \, \mathrm{d}\, q.$$

The LHS can be interpreted as the difference between the percentage of subscription in two markets as if the log of consumption ($\tilde{Q}_{it,m}$) was uniformly distributed. The moment equation links the change of subscription percentage to the change of price across markets.

5 Empirical application: music streaming service subscription

In this empirical application, we focus on the market of online music streaming service in Taiwan during the period January 2015–February 2017. We examine an empirical setting in which we study the churn decision of a customer, and we use our method to obtain estimates of the elasticities of the WTP to age and gender, the mean and CDF of the WTP for the monthly streaming service and a counterfactual weekly service.

5.1 Data

The data were provided by a music streaming service company, which targets Southeast Asian markets, including Taiwan (China), Hong Kong (China), Malaysia, Japan etc. Its service has 80 percent market share in the Taiwan, a province of China. Though the company sells subscription plans of varying lengths (e.g., Monthly, 180 days, 365 days), most users (93.7 percent in our sample) choose monthly plan. Registered users can also listen to music free for up to 1 hour each day with various restrictions, however only 3.56 percent of the users in our sample have ever used this free service. We will focus on the subscription choice of monthly plan, which has three different prices in our sample. The first price is its listed price 149 New Taiwan dollar (NT$), which equals 4.73 USD. The second price is 129 NT$, which is only available to the customers of certain cellular carriers in Taiwan. The last price is 99 NT$, which is the introductory offer for new users who paid with VISA card. This introductory offer is only valid for the first two months. In our sample, the percentages of the three prices 149, 129, and 99 are 56.8, 17.7 and 23.9, respectively. It should be remarked that these prices are for the same monthly music streaming service.

Img/AverageDailyHoursChurnVSNotchurn.png

Figure 1: Kernel Density Functions of (Log) Average Monthly Listening Hours

The raw data include more than 1 million registered users in Taiwan. In data, we observe the daily usage (the daily number of seconds each user listened music with the service) of its subscribers from January 1st 2015 to February 28th 2017. We also observe each user's transaction history during that period. So we know which users have churned, or left the service. We define that a customer churns if there is no new valid service subscription within 30 days after the his or her current membership expires. According to this definition, only 2.5 percent of customers who have churned would come back to use the service some time later. As for user's characteristics, we can observe age, gender and city. It turns out that city does not affect usage and the WTP significantly, hence we can will only use age and gender as the observed characteristics. Also, the age and gender of many users are missing. Comparing the kernel densities of the usage by those with and without age and sex information, we did not find significant difference, hence we removed the user without valid age and gender observations. We randomly sampled 8698 users for the analysis below.

?? shows the basic summary statistics of the market by summarizing customers' usage and characteristics. From this table, one can see that there is no significant difference in term of age and gender between the customers who churned and who did not. Looking at their daily usage, customers who churned used less than those who did not. **??** further illustrates this observation by plotting the kernel density functions of the average and the log of the average monthly listening hours.

5.2 Model

Let S_i denote customer i's churn decision, and $S_i = 0$ if he or she churns and $S_i = 1$ otherwise. Her choice follows

$$S_i = \mathbb{I}(\ln W_i > \ln(P_i + \mu)),$$

Table 2: Descriptive Statistics of Music Streaming Service Sample

Panel A: Users who churned ($n = 1211$)						
	Mean and std. dev.	First Quartile	Median	Third Quartile	Min	Max
Average daily listening hours (lifetime in sample)	1.17 (0.99)	0.45	0.88	1.61	0.02	6.69
Tenure (weeks)	139.82 (93.06)	55.93	130.57	210.93	0.14	349.57
Age	29.53 (8.25)	23.00	28.00	34.00	17.00	59.00
Is male	0.51					
Panel B: Users who did not churn ($n = 7487$)						
	Mean and std. dev.	First Quartile	Median	Third Quartile	Min	Max
Average daily listening hours (lifetime in sample)	1.33 (1.09)	0.55	1.04	1.81	0.02	6.74
Tenure (weeks)	164.40 (99.89)	68.43	162.86	252.43	0.14	376.86
Age	29.92 (8.14)	24.00	28.00	34.00	17.00	59.00
Is male	0.55					

where W_i is her WTP for the monthly music streaming service, and $P_i \in \{99 \text{ NT\$}, 129 \text{ NT\$}, 149 \text{ NT\$}\}$ is her price, and μ is the money-metric expected utility from outside option. We keep using the multiplicative specification of the WTP $W_i = \alpha_i Q_i^*$ or

$$\ln W_i = \ln \alpha_i + \ln Q_i^* = \beta_1 + \beta_2 Age_i + \beta_3 Female_i + \ln Q_i^* + U_i,$$

where $Female_i$ is a female dummy variable, and Q_i^* is the expected monthly usage. For users who churn, we let Q_i^* be the total listening hours during the 30 days before he or she churn. For users who did not churn, we let Q_i^* be the average monthly listening hours in their subscription period.

Our estimation follows the following steps. First, we use kernel density estimator to estimate the conditional PDF $f_{\ln Q^*}(\ln Q_{i,}^* \mid Age_{i,}, Female_{i,})$. Second, because we have three prices in data, we use the moment **??** to estimate the money-metric expected utility μ of the outside option. In particular, we find the minimizer of

$$\hat{\mu} = \arg\min_{\mu > 0} \hat{g}(\mu)' \hat{g}(\mu),$$

where $\hat{g}(\mu) = (\hat{g}_{129}(\mu), \hat{g}_{99}(\mu))'$. For $p = 99, 129, 149$, let

$$f_{i,p} = f_{\ln Q^*}(\ln Q_{i,p}^* \mid Age_{i,p}, Female_{i,p}),$$

the moment function $\hat{g}_p(\mu)$ is defined as follows,

$$\hat{g}_p(\mu) = \left[\frac{1}{n_p} \sum_{(i,p)} \frac{S_{i,p}}{f_{i,p}} - \frac{1}{n_{149}} \sum_{(i,149)} \frac{S_{i,149}}{f_{i,149}}\right] - \left[\frac{1}{n_p} \sum_{(i,p)} \frac{\mathbb{I}(Q_{i,p}^* - \ln(p+\mu) > 0)}{f_{i,p}} - \frac{1}{n_{149}} \sum_{(i,149)} \frac{\mathbb{I}(Q_{i,149}^* - \ln(149+\mu) > 0)}{f_{i,149}}\right],$$

where (i, p) denotes a customer who paid p for his or her monthly plan, and n_p is the number of consumers who paid p for their monthly plan in the sample. The estimated money-metric expected utility of outside option is $\hat{\mu} = 38.15$ with standard error 9.24. The standard error was obtained from bootstrap. Third, using the data of subscribers who paid 149 NT\$ for their monthly plan, we estimate $\beta = (\beta_1, \beta_2, \beta_3)'$ by linear regression of

$$Y_{i,149} = \frac{S_{i,149} - \mathbb{I}(\ln Q_{i,149}^* - \ln(149 + \hat{\mu}) > 0)}{f_{\ln Q^*}(\ln Q_{i,149}^* \mid Age_{i,149}, Female_{i,149})},$$

on $Age_{i,149}$ and $Female_{i,149}$. Fourth, we estimate nonparametrically

$$\pi(Age_{i,149}, Female_{i,149}, \ln Q_{i,149}^*) \equiv \mathrm{E}(S_{i,149} \mid Age_{i,149}, Female_{i,149}, \ln Q_{i,149}^*).$$

Then the CDF of the WTP is obtained from the formula in **??**.

Table 3: WTP and Usage Elasticities (Hundredths) Estimates

	WTP	Usage (All Users)	Usage (Churn)	Usage (Not Churn)
Age	3.98	−2.00	−2.18	−2.09
	(1.19)	(0.22)	(0.65)	(0.20)
Is female	−26.23	−3.13	9.49	−4.75
	(18.09)	(3.46)	(10.34)	(3.14)
n of obs.	4159	4159	913	3246

5.3 Results

?? reports the estimated elasticities of the WTP and usage to age and gender using **??**, and by approximating $\mathrm{E}(\ln Q_i^* \mid Age_i, Female_i)$ with a linear regression of $\ln Q_i^*$ on Age_i and $Female_i$. It is interesting to see that even though age elasticity of the usage is negative, age elasticity of the WTP is significantly positive. This is likely due to the fact that age is positively correlated with earnings, and an elder user is willing to pay more for one unit of usage (e.g., one hour music). Their WTP for listening one hour of music is high enough so that the WTP for the monthly service increases despite their lower usage than younger users. The effects of gender on the WTP is insignificant (with p-value 0.13), though the sign of the elasticity may be due to different earnings of men and women.

?? reports the estimated mean of log of WTP $\mathrm{E}(\ln W_i \mid Age_i, Female_i)$ and mean of log monthly number of listening hours. We divided age into four categories, and estimated the mean for each category and sex. We observe that on average, one is willing to pay substantially more than the current listed price \$4.7 regardless of age and gender. This explains the very high renewal rate (more than 93 percent in sample) in a given month and its high market share. We do not observe gender difference in the WTP for the age before college. This is likely because these high school students have not started earning any income, so gender earnings gap does not affect the WTP. For the age between 19 and 30, men's average WTP is higher than women's. This is not true when age exceed 30, where women's average WTP is greater than men's because men on average listens less music than women after age 30.

?? shows the estimated CDF of the WTP of a man when his usage at 10 and 50 percentile of the population. Looking at the median, one's WTP is very elastic to his usage.

?? shows the estimated demand curve. The y-axis is the counterfactual price, and the x-axis is the percentage of the current customers who would pay such counterfactual price and subscribe the monthly service. It is estimated by using **??**. The estimated curve suggests

Table 4: Log of WTP and Usage Estimates

	Mean of Log WTP		Mean of Log Monthly Hours	
	Men	Women	Men	Women
Age $\leq$ 18 (before college)	log($16)	log($16)	log(26.6 Hrs)	log(25.9 Hrs)
Age between 19 and 22 (college)	log($24)	log($21)	log(31.9 Hrs)	log(27.9 Hrs)
Age between 23 and 30	log($33)	log($30)	log(29.4 Hrs)	log(26.7 Hrs)
Age $>$ 30	log($49)	log($56)	log(19.6 Hrs)	log(21.5 Hrs)

Note: The listed monthly subscription price is $4.7.

Img/CDFofWTPMan.png

Figure 2: Estimated CDF of the WTP of Men

Img/DemandCurve.png

Figure 3: Estimated Demand Curve

that consumers are very inelastic at the current price. One caveat is that in estimating the demand curve, we rely on

$$\Pr(S_i^c = 1 \mid X_{1i}, q_i, P^c) = \pi_1(X_{1i}, \ln(P + \mu) - \ln(P^c + \mu) + q_i),$$

where the superscript c denotes a counterfactual object. Note that $\Pr(S_i^c = 1 \mid X_{1i}, q_i, P^c)$ is estimable only when $\ln(P + \mu) - \ln(P^c + \mu) + q_i$ is within the support of observed log of usage in data (which is generated under the current price). When the counterfactual price P^c differs greatly from the current price, such a support condition may not hold and the obtained demand $\Pr(S_i^c = 1 \mid P^c)$ is unreliable. This implies that one may not be able to identify the optimal price when the usage variation is not big enough, but the firm can at least identify whether or not the current price is too high or low because the slope of the profit function at the current price is identified when the marginal cost is known.

?? shows the CDF of the WTP for a counterfactual weekly music streaming service for different age group. This CDF was derived based on the multiplicative specification of the WTP. Denote Q_i^w the weekly usage (listening hours) by a customer i. By the multiplicative specification, the log of the WTP, $\ln W_i^w$, equals $\beta_1 + \beta_2 Age_i + \beta_3 Female_i + \ln Q_i^w + U_i$. Given a counterfactual price P^{cw} for the counterfactual weekly plan, it is easy to see from **??** that

$$F(W_i^w \mid Age_i, Female_i, Q_i^w) = 1 - \pi_1(Age_i, Female_i, \ln(P^{cw} + \mu^w) - \ln W_i^w + \ln Q_i^w),$$

where μ^w is the counterfactual money-metric expected utility of the outside option for one week (e.g. let $\mu^w = \mu/4$). Then we have

$$F(W_i^w \mid Age_i, Female_i) = 1 - \mathrm{E}(\pi_1(Age_i, Female_i, \ln(P^{cw} + \mu^w) - \ln W_i^w + Q_i^w)),$$

Img/WeeklyPlanWTP.png

Figure 4: CDF of the WTP for a Counterfactual Weekly Subscription Plan

where is estimable because we know the distribution weekly log usage Q_i^w from the daily usage tracking data.

6 Conclusion

Many subscription commerce markets charge the same price for every consumer. Price variation is very limited, if there is any. Classic results and arguments from the literature discuss how the identification of demand or WTP is not possible without price variation.

This paper shows that if one is willing to impose certain functional form of the WTP as a function of the usage, big usage tracking data and observed subscription choices can identify the elasticities and the distribution function of the WTP. Crucially, our approach works because purchase (subscription) is separated from usage, and the two are related in the sense that obtaining a subscription opens up for the consumer the possibility of using the service for a potentially unlimited amount.

However, without functional form restriction, we find that usage tracking data alone cannot identify the elasticities or the distribution of the WTP. However, price variation, even in limited, can help identify the functional form of the WTP.

A Proofs

For simplicity, we omit the subscript i or (i,t) in the proof below whenever there is no confusion.

Proof of **??** *and* **??**. First, we get the formula of β. Letting

$$\tilde{X}_1' = Z'[\mathrm{E}(ZZ')]^{-1}\,\mathrm{E}(ZX_1'),$$

it is known that (**?**)

$$\mathrm{E}(\tilde{X}_1 X_1')\beta = \mathrm{E}(\tilde{X}_1 Y_1),$$

hence

$$\begin{aligned}\beta &= [\mathrm{E}(X_1 Z')[\mathrm{E}(ZZ')]^{-1}\,\mathrm{E}(ZX_1')]^{-1}[\mathrm{E}(X_1 Z')[\mathrm{E}(ZZ')]^{-1}\,\mathrm{E}(ZY_1)] \\ &= [\mathrm{E}(\tilde{X}_1\tilde{X}_1')]^{-1}\,\mathrm{E}(\tilde{X}_1 Y_1).\end{aligned}$$

Second, we obtain the formula of $F_U(U \mid X_1)$. Let

$$\pi_1(x_1, z, q) = \mathrm{E}(S \mid X_1 = x_1, Z = z, \ln Q^* = q).$$

We have

$$\begin{aligned}\pi_1(x_1, Z, q) &= \Pr(U > -(\beta' x_1 + q - \ln P) \mid X_1 = x_1, Z, \ln Q^* = q) \\ &= \Pr(U > -(\beta' x_1 + q - \ln P) \mid X_1 = x_1, Z) \\ &= 1 - F_U(\ln P - \beta' x_1 - q \mid X_1 = x_1, Z).\end{aligned}$$

In other words,

$$F_U(u \mid X_1, Z) = 1 - \pi_1(X_1, Z, \ln P - \beta' X_1 - u). \tag{A.1}$$

Third, we obtain the formula of $F_W(W \mid X_1, Z, Q^*)$. We have

$$\begin{aligned}F_W(w \mid X_1 = x_1, Z, Q^* = e^q) &= \Pr(\ln W \le \ln w \mid X_1 = x_1, Z, \ln Q^* = q) \\ &= \Pr(\beta' x_1 + U + q \le \ln w \mid X_1 = x_1, Z, \ln Q^* = q) \\ &= \Pr(U \le \ln w - \beta' x_1 - q \mid X_1 = x_1, Z, \ln Q^* = q) \\ &= F_U(\ln w - \beta' x_1 - q \mid X_1 = x_1, Z) \\ &= 1 - \pi_1(x_1, Z, \ln P - \ln w + q).\end{aligned}$$

Note that the above lines also establish $F_W(w \mid X_1, Z, Q^*) = F_U(\ln w - \beta' X_1 - \ln Q^* \mid X_1, Z)$.

Fourth, we derive the formula of $\mathrm{E}(\ln W \mid X_1)$. We start with deriving the formula of $\mathrm{E}(\ln W \mid X_1, Z, Q^*)$. By our specification, we have

$$\ln W = \beta' X_1 + \ln Q^* + U.$$

Because $Q^* \perp\!\!\!\perp U \mid (X_1, Z)$, we have

$$\mathrm{E}(\ln W \mid X_1, Z, Q^*) = \beta' X_1 + \ln Q^* + \mathrm{E}(U \mid X_1, Z).$$

Note that $\mathrm{E}(U \mid X_1, Z) \neq 0$ when X_1 is correlated with U. Below, we will show that

$$\mathrm{E}(U \mid X_1, Z) = \mathrm{E}(H_1 \mid X_1, Z). \tag{A.2}$$

We then have

$$\mathrm{E}(\ln W \mid X_1) = \beta' X_1 + \mathrm{E}(\ln Q^* \mid X_1) + \mathrm{E}(H_1 \mid X_1).$$

We now show **??**. First, it follows from the integration by parts that

$$\mathrm{E}(U \mid X_1, Z) = \int_{-\infty}^{\infty} [\mathbb{I}(u > 0) - F_U(u \mid X_1, Z)] \, \mathrm{d}\, u.$$

Using the identified formula of $F(u \mid X_1, Z)$, we have

$$\mathrm{E}(U \mid X_1, Z) = \int_{-\infty}^{\infty} [\mathrm{E}(S \mid X_1, Z, \ln Q^* = \ln P - \beta' X_1 - u) - \mathbb{I}(u \leq 0)] \, \mathrm{d}\, u.$$

Using the substitution of variables, we have

$$\begin{aligned}
\mathrm{E}(U \mid X_1, Z) &= \int_{-\infty}^{\infty} [\mathrm{E}(S \mid X_1, Z, \ln Q^*) - \mathbb{I}(\ln P - \beta' X_1 - \ln Q^* \leq 0)] \, \mathrm{d}(\ln Q^*) \\
&= \mathrm{E}\left(\frac{S - \mathbb{I}(\ln P - \beta' X_1 - \ln Q^* \leq 0)}{f_{\ln Q^*}(\ln Q^* \mid X_1, Z)} \,\middle|\, X_1, Z \right) \\
&= \mathrm{E}(H_1 \mid X_1, Z),
\end{aligned}$$

and that is **??**. □

Proof of **??**. For simplicity, denote $\ln Q^*$ by q in the proof below. We prove **??** in two cases. In the first case, $U < 0$, and in the second case, $U \geq 0$.

Case 1: $U < 0$**.** We have

$$\begin{aligned}
\lambda &= \int_{-\infty}^{\ln P - U} \mathbb{I}(q > \ln P - U) - \mathbb{I}(q > \ln P) \, \mathrm{d}\, q + \\
&\qquad \int_{\ln P - U}^{\infty} \mathbb{I}(q > \ln P - U) - \mathbb{I}(q > \ln P) \, \mathrm{d}\, q \\
&= \int_{-\infty}^{\ln P - U} 0 - \mathbb{I}(q > \ln P) \, \mathrm{d}\, q + 0 \\
&= -\int_{\ln P}^{\ln P - U} \mathrm{d}\, q = U.
\end{aligned}$$

Case 2: $U \geq 0$**.** We have that

$$\begin{aligned}
\lambda &= \int_{-\infty}^{\ln P - U} \mathbb{I}(q > \ln P - U) - \mathbb{I}(q > \ln P)\ \mathrm{d}\, q + \\
&\qquad \int_{\ln P - U}^{\infty} \mathbb{I}(q > \ln P - U) - \mathbb{I}(q > \ln P)\ \mathrm{d}\, q \\
&= 0 + \int_{\ln P - U}^{\infty} \mathbb{I}(q > \ln P - U) - \mathbb{I}(q > \ln P)\ \mathrm{d}\, q \\
&= \int_{\ln P - U}^{\ln P} \mathbb{I}(q > \ln P - U) - 0\ \mathrm{d}\, q \\
&= \int_{\ln P - U}^{\ln P} \mathrm{d}\, q = U.
\end{aligned}$$

We then have shown **??**. □

Proof of **??** *and* **??**. First, we get the formula of β. It is known from **?** that

$$\mathrm{E}(\tilde{X}_1 X_1')\beta = \mathrm{E}(\tilde{X}_1 Y_2).$$

Hence

$$\beta = [\mathrm{E}(\tilde{X}_1 \tilde{X}_1')]^{-1}\, \mathrm{E}(\tilde{X}_1 Y_2).$$

Second, let

$$\pi_2(x_1, z, q) = \mathrm{E}(S \mid X_1 = x_1, Z = z, \tilde{Q} = q),$$

and we have

$$F_\eta(\eta \mid X_1, Z) = 1 - \pi_2(X_1, Z, \ln P - \beta' X_1 - \eta)$$

by similar arguments in the proof of **??**.

Third, we obtain the formula of $F_W(w \mid X, Z)$. By similar arguments, we have

$$F_W(w \mid X, Z) = 1 - \pi_2(X_1, Z, \ln P - \ln w + \gamma' X_2).$$

Using the notation of $\tilde{Q} = \gamma' X_2$, we write $\ln W = \beta' X_1 + \tilde{Q} + \eta$, hence

$$\mathrm{E}(\ln W \mid X_1, X_2, Z) = \beta' X_1 + \gamma' X_2 + \mathrm{E}(\eta \mid X_1, X_2, Z).$$

When X is correlated with either U or V, $\mathrm{E}(\eta | X, Z) \neq 0$. By the law of iterated expectation,

$$\mathrm{E}(\ln W \mid X_1, X_2) = \beta' X_1 + \gamma' X_2 + \mathrm{E}(\eta \mid X_1, X_2).$$

Below we will show that

$$\mathrm{E}(\eta \mid X_1, X_2) = \mathrm{E}(H_2 \mid X_1, X_2),$$

where H_2 is as defined in **??**. We first have $\mathrm{E}(\eta \,|\, X_1, X_2, Z) = \mathrm{E}(\eta \,|\, X_1, Z)$ by the conditional independence assumption, thus

$$\begin{aligned}\mathrm{E}(\eta \mid X_1, X_2) &= \mathrm{E}(\mathrm{E}(\eta \mid X_1, X_2, Z) \mid X_1, X_2) \\ &= \mathrm{E}(\mathrm{E}(\eta \mid X_1, Z) \mid X_1, X_2).\end{aligned}$$

Using the similar arguments in the proof of **??**, we can show that

$$\mathrm{E}(\eta \mid X_1, Z) = \mathrm{E}\left(\frac{S - \mathbb{I}(\ln P - \beta' X_1 - \tilde{Q} \leq 0)}{f_{\tilde{Q}}(\tilde{Q} \mid X_1, Z)} \,\middle|\, X_1, Z \right) = H_2.$$

□

B Numerical studies

We show numerical performance of the proposed method here. We generate data from the following model,

$$\begin{aligned}S_{it} &= \mathbb{I}(\beta_1 + \beta_2 X_{i1t} + \gamma_1 + \gamma_2 X_{2it} - \ln P + V_i + U_{it} > 0) \\ &= \mathbb{I}(0.5 + 1 \cdot X_{i1t} + 0.5 + 1 \cdot X_{2it} - \ln P + V_i + U_{it} > 0), \\ Q_{it} &= \gamma_1 + \gamma_2 X_{2it} + V_i + \varepsilon_{it},\end{aligned}$$

where

$$\begin{aligned}&V_i = \omega_{1i}, && U_{it} = \omega_{1i} + \omega_{2it}, && X_{i1t} = \omega_{1i} + \omega_{3it}, \\ &Z_{it} = (1, \omega_{3it})', && X_{2it} = \omega_{4it}, && \varepsilon_{it} = \omega_{5it}.\end{aligned}$$

Here ω's are mutually independent normally distributed random variables. So U_{it} and V_i are correlated, X_{i1t} is endogenous (correlated with both V_i and U_{it}), Z_{it} is IV for X_{i1t}, and X_{2it} is exogenous whose support conditional on (X_{i1t}, Z_{it}) is real line. We assume that $\omega_{1i} \sim \mathcal{N}(0, 0.5^2)$, $\omega_{2it} \sim \mathcal{N}(0, 0.5^2)$, $\omega_{3it} \sim \mathcal{N}(1, 1)$, $\omega_{4it} \sim \mathcal{N}(1, \sigma_{x_2}^2)$, $\omega_{5it} \sim \mathcal{N}(0, 1)$. By this construction, $\eta_{it} = U_{it} + V_i$ and $\eta_{it} \sim \mathcal{N}(0, 1.25)$. It is easy to show that $\eta_{it} \mid X_{i1t} = x_1 \sim \mathcal{N}(0.4(x_1 - 1), 1.05)$. Hence the $W_{it} \,|\, (X_{i1t} = x_1, X_{2it} = x_2)$ follows a log normal distribution, and the mean and variance of its logarithm are $1.4x_1 + x_2 + 0.6$ and 1.05, respectively. We let $\ln P = 2$.

One important condition is that the support of $\tilde{Q}_{it,m} \mid X_{i1t}$ covers the support of $\ln P - \beta' X_{i1t} - \eta_{it} \mid X_{i1t}$. In this numerical example, this condition requires that the support of $X_{2it} \sim \mathcal{N}(1, \sigma_{x_2}^2)$ covers the support of $1 - X_{i1t} - \eta_{it} \,|\, X_{i1t} \sim \mathcal{N}(1.4(1 - X_{i1t}), 1.05)$. Whether or not this support condition holds depends on the value of X_{i1t} and σ_{x_2}. **??** displays the

Figure 5: Illustrate Support Condition

Img/SupportCond.png

95% confidence interval as well as the density function of X_{2it} and $1 - X_{i1t} - \eta_{it} \mid X_{i1t}$ when $\sigma_{x_2} = 0.5$, 1 and 1.5, and X_{i1t} equals to 10%, 25%, 50%, 75% and 90% percentile of its marginal distribution. It is clear that when σ_{x_2} is too small ($\sigma_{x_2} = 0.5$), the support hardly hold. When we increase σ_{x_2}, it is easier to satisfy the support condition.

All reported results were based on 250 replications. **??** shows the estimation of β_2, γ_2 and $\beta_1 + \gamma_1$. Note that we cannot separately identify β_1 and γ_1. It is interesting to observe that when σ_{x_2} is small making the support condition harder to withstand, there is significant bias in estimating β_2. It is also interesting to observe that when $\sigma_{x_2} = 1.5$ and the support condition does not strictly hold (see **??**), the bias in estimating β has became very small. The effects of support conditions are further illustrated by the CDF estimation in **??** and the estimation of $\mathrm{E}(\ln W_{it} \mid X_{i1t}, X_{2it})$ in **??**. Comparing the bias of estimating $\mathrm{E}(\ln W_{it} \mid X_{i1t}, X_{2it})$ in **??** with the support condition **??**, one can see that the bias becomes larger when it becomes more harder to satisfy the support condition.

Table 5: Parameter Estimates: $T = 2$

	$n = 1000$			$n = 2000$		
	$\beta_2 = 1$	$\gamma_2 = 1$	$\beta_1 + \gamma_1 = 1$	$\beta_2 = 1$	$\gamma_2 = 1$	$\beta_1 + \gamma_1 = 1$
$\sigma_{x_2} = 0.5$	0.551	1.004	1.005	0.529	0.992	1.029
	(0.159)	(0.086)	(0.181)	(0.124)	(0.065)	(0.148)
$\sigma_{x_2} = 1.0$	0.824	0.999	1.074	0.846	0.999	1.061
	(0.181)	(0.048)	(0.145)	(0.160)	(0.034)	(0.180)
$\sigma_{x_2} = 1.5$	0.962	1.002	1.048	0.962	1.000	1.066
	(0.208)	(0.036)	(0.253)	(0.162)	(0.023)	(0.100)

Note: Results are based on 250 replications, and the standard deviation is in parenthesis.

Figure 6: Estimation of the CDF of $\eta \mid X_{i1t} = 1$ and $W_{it} \mid X_{i1t} = 1, X_{2it} = 1$

Img/cdf_n_1000.png

Figure 7: Estimation of $\mathrm{E}(\ln W_{it} \mid X_{i1t}, X_{2it})$

Img/mean_n_1000.png

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